Fantasy Football Expert Narratives: Signal or Storytelling?

This reading gives you the context and technical tools to audit whether fantasy football expert analyses are actually useful for predicting player performance.

What is fantasy football?

Fantasy football is a management simulation game where participants manage fake NFL rosters. At its core, you act as the General Manager and Head Coach of a virtual roster composed of real-life NFL athletes. Your success depends on your ability to predict human performance using historical data, matchups, and injury reports.

The Draft

The season begins with the Draft where participants take turns selecting players from the entire pool of NFL athletes. Once a player is drafted in your league, they are off the board—no two teams in the same league can own the same player. This creates a market where you must balance the best player available with your specific roster needs.

Rosters

A standard fantasy roster mimics a real offensive unit, typically requiring you to start:

- 1 Quarterback (QB)
- 2 Running Backs (RB)
- 2-3 Wide Receivers (WR)
- 1 Tight End (TE)
- 1 Kicker (K)
- 1 Defense and Special Teams Unit (D/ST)

Scoring

Fantasy football converts real-world physical achievements into a singular mathematical score. While settings vary, the standard logic remains:

- Yardage: Players earn points for every 10 yards gained rushing/receiving or 25 yards passing.
- Touchdowns: The highest-value events (usually 4–6 points).
- Receptions: Many leagues use PPR (Points Per Reception), awarding a point simply for catching the ball, which adds a layer of value to high-volume receivers.
- Negatives: Fumbles, interceptions, and missed kicks subtract from your total.

Team Management

Once the season starts, your role shifts to a team manager. You must navigate:

- Weekly Matchups: You go head-to-head against one other manager in your league. The team with the most points at the end of Monday Night Football wins the week.
- The Waiver Wire: Not every impact player is drafted. The waiver wire is the "free agent" pool where you can "claim" players who are performing well or replacing injured starters.
- Trades: Managers can negotiate deals to swap players, often trading depth at one position (like RB) to upgrade a weakness at another (like WR).
- Start/Sit Decisions: Every week, you must decide which players move to your "Bench" and which stay in your "Starting Lineup." Points scored by players on your bench do not count toward your weekly total.

This is where the "expert" fantasy analyst industry was born. Because NFL performance is volatile, every decision is a gamble on the outcome of an event. You are trying to identify who has the highest ceiling and the safest floor in a specific window of time. This is why data analysis is so vital.

<u>What articles am I reviewing?</u>
Analyst outlook articles are weekly fantasy media pieces such as:

- Start/Sit recommendations
- Waiver-wire targets
- Matchup-based rankings
- “Boom or bust” profiles

These pieces are usually written in persuasive language (“must start,” “high-upside,” “avoid this week”). That tone can shape reader behavior, even when the evidence behind the recommendation is uncertain. You are auditing whether the tone of these recommendations helps forecast performance, or mainly reflects narrative framing.

What is sentiment analysis?

Sentiment analysis converts text tone into numeric values:

- Positive tone (optimistic language)
- Neutral tone
- Negative tone (pessimistic language)

In this project, sentiment scores are extracted from fantasy articles and linked to player-week outcomes. If sentiment is predictive, more positive writeups should generally align with stronger future fantasy outcomes. The core question is whether sentiment adds genuine signal beyond what is already known from stats and public context.

Fantasy media is influential, but influence does not equal accuracy. A positive article may coincide with a strong week for two very different reasons:

1. True predictive signal: analyst language captures real future value.

2. Narrative echo: analyst language just reflects obvious recent stats, bias, or hype.

If #2 dominates, sentiment can appear useful while adding little new information. Analysts often react to:

- Last week's breakout game (recency bias)
- Existing star reputation (popularity bias)
- Injury news and team narratives
- Consensus rankings already available elsewhere

In these cases, positive tone may track what already happened, not what will happen next. The goal is to determine whether analyst narratives are generally reliable, selectively useful, or mostly storytelling.

Technical Reading

***Important Note:** For this case study, sentiment scores have already been calculated from text_dataset.json. This study focuses mainly on extracting insights from these scores and examining the correlation with player statistics.*

This section should is a guided walkthrough of how the sentiment analysis was built and how the first exploratory checks were done. It provides context for how the project turns fantasy article text into a numeric feature and how that feature can be compared with player performance.

Sentiment analysis basics

Sentiment analysis is a way of turning text into numbers. A sentiment model maps words/phrases to positivity or negativity and combines them into a score. In this project, the key metric is a compound score from -1 to +1.

- Near +1: strongly positive tone
- Near 0: neutral/mixed tone
- Near -1: strongly negative tone

For the case study, that means we can treat analyst outlooks as data. If fantasy writers are truly helping forecast performance, then stronger positive sentiment might line up with better fantasy results. If not, the tone may be mostly storytelling, recency bias, or general enthusiasm.

Exploratory Data Analysis

A good first step is to inspect the sentiment distribution in the final dataset. First, read the csv and prin the first few lines.

```
df = pd.read_csv("fantasy_dataset.csv")
print(df.head())

week       player_name  sentiment_compound  sentiment_pos  sentiment_neg  \
0    18        Mike Evans              0.6988          0.092          0.069
1    18     George Kittle              0.8740          0.108          0.053
2    18      Nico Collins              0.5429          0.056          0.022
```

Next, we want to get an overview of the sentiment scores in our dataset. To do this, define our target variable as a player + week record, and look at the distribution of sentiment scores.

```
player_week = (
   df.groupby(["week", "player_name", "position", "Team"])
     .agg(
         mean_sentiment=("sentiment_compound", "mean"),
         sentiment_std=("sentiment_compound", "std"),
         article_count=("sentiment_compound", "count"),
         mean_word_count=("word_count", "mean"),
         mean_pos=("sentiment_pos", "mean"),
         mean_neg=("sentiment_neg", "mean"),
         mean_neu=("sentiment_neu", "mean"),
         TotalPoints=("TotalPoints", "first"),
         Rank=("Rank", "first")
     )
     .reset_index()
)
print(player_week.describe())
       mean_sentiment  sentiment_std  article_count  \
count    2000.000000    2000.000000    2000.000000
mean     0.396167       0.173630       1.909500
std      0.533662       0.299197       1.627462
```

This first check gives you a basic sense of the tone used in fantasy media. In this project, the average sentiment appears positive overall, which suggests that fantasy articles often lean optimistic. That is not surprising, since analysts are usually trying to recommend playable options rather than discourage readers. But it also raises an important question: if most of the writing is already positive, does that positivity actually help predict fantasy output?

The next step is to inspect the raw article text from text_dataset.json. This project does not only use the final numeric dataset; it also preserves the article-level text to help understand how sentiment is being generated. We want to examine the article metadata to get a broader view of our dataset. These checks matter because they help explain how the text data was organized before sentiment scoring.

For example, we can check out the unique player names from our sentiment analyses:

```
with open("Data/text_dataset.json", "r", encoding="utf-8") as f:
       data = json.load(f)

# Use a set to automatically handle uniqueness
unique_names = set()

for article in data:
    for player in article.get("players", []):
        name = player.get("name", "").strip()
        if name:
            unique_names.add(name)

# Sort alphabetically for easier reading
sorted_names = sorted(list(unique_names))

print(f"--- Player List Audit ---")
print(f"Total Unique Names Found: {len(sorted_names)}")
print("-" * 30)

for name in sorted_names:
    print(name)
```

We may also want to generate exploratory plots from our sentiment dataset. First, load the dataset.

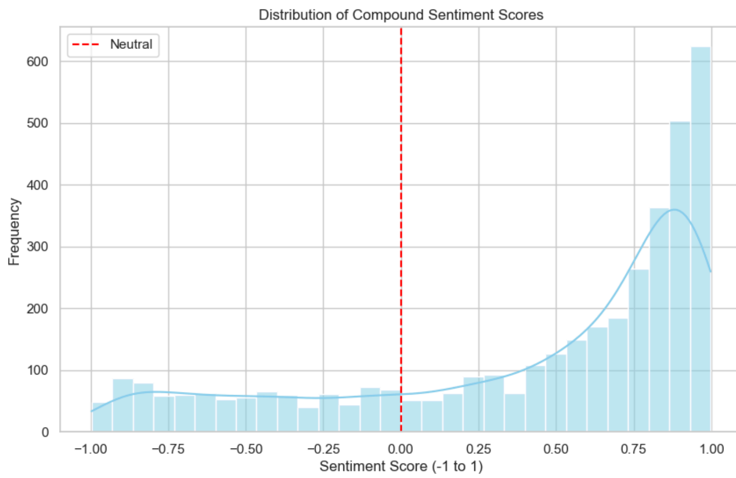
```
# Load the data
df = pd.read_csv('../DATA/fantasy_dataset_final.csv')

# Drop rows where we don't have fantasy points (for correlation plots)
df_clean = df.dropna(subset=['TotalPoints'])
```

Next, use seaborn and matplotlib to generate exploratory plots. For example, we can generate the distribution of sentiment scores.

```
# Plot 1: Distribution of Sentiment Scores
# Goal: See if analysts are generally positive, negative, or neutral.
plt.figure(figsize=(10, 6))
sns.histplot(df['sentiment_compound'], bins=30, kde=True, color='skyblue')
plt.title('Distribution of Compound Sentiment Scores')
plt.xlabel('Sentiment Score (-1 to 1)')
```

```
plt.ylabel('Frequency')
plt.axvline(0, color='red', linestyle='--', label='Neutral')
plt.legend()
plt.show()
```



We can also generate a scatterplot comparing sentiment with total points. This is a helpful first look because it shows whether more positive language appears alongside better fantasy outcomes. It is important to note that these plots do not prove prediction. They only help the see whether there is an obvious pattern that may appear once we investigate further.

```
sns.scatterplot(data=df_clean, x='sentiment_compound', y='TotalPoints',
hue='position', alpha=0.6, ax=axes[1])
sns.regplot(data=df_clean, x='sentiment_compound', y='TotalPoints', scatter=False,
color='black', ax=axes[1])

axes[1].set_title('Sentiment Score vs. Total Fantasy Points')
axes[1].set_xlabel('Sentiment Compound Score (-1 to 1)')
axes[1].set_ylabel('Total Points Scored')

plt.tight_layout()
plt.show()
```

A deeper EDA is provided in the Technical References directory of this case study. This may b eused as an initial reference point for generating the report on the effectiveness of article sentiment in predicting player performance.

There are some things that must be taken into consideration. For example, some positions may receive more positive framing, more caution, or more disagreement than others. Also, different positions may have different average or median total points. These exploratory plots do not answer the final question, but they help identify that sentiment AND performance may not be uniform across the NFL.

Analysis Approach

The question to answer is: are higher sentiment values associated with better future fantasy outcomes? Below are some initial ideas and things to consider when attemmpting to answer this question. You should use this example and build off of it, diving deeper into the player outlooks.

Testing Whether Sentiment Actually Matters

At this point, you have looked at the shape of the data, the distribution of sentiment, and a few basic relationships between tone and fantasy points. That is a good start, but it is not enough to answer the hook. To get closer to the real question you need to move into statistical testing and modeling.

The first thing I would do is establish a baseline. Before adding sentiment, ask how much fantasy performance can be explained by obvious football variables like position. That gives you something to compare against.

```
import pandas as pd
import statsmodels.formula.api as smf

df = pd.read_csv("../DATA/fantasy_dataset_final.csv")

# Baseline model: position only
baseline = smf.ols("TotalPoints ~ C(position)", data=df).fit()
print(baseline.summary())
```

This matters because if position already explains most of the variation, sentiment has to add something beyond that to be useful. A model that only repeats known role differences is not really helping the fantasy manager much.

Next, add sentiment to the model. This is the first real test of the hook question.

```
model1 = smf.ols(
   "TotalPoints ~ sentiment_compound + C(position)",
   data=df
).fit()

print(model1.summary())
```

If sentiment_compound is not significant here, that suggests tone alone is not doing much after accounting for basic player role. If it is significant, that is still not the end of the story it could be picking up something else, like player popularity or matchup strength.

A better version of this analysis is to use the player-week dataset, since that reduces duplicate observations and gives a cleaner unit of analysis.

```
player_week = (
   df.groupby(["week", "player_name", "position", "Team"])
     .agg(
         mean_sentiment=("sentiment_compound", "mean"),
         sentiment_std=("sentiment_compound", "std"),
         article_count=("sentiment_compound", "count"),
         TotalPoints=("TotalPoints", "first"),
         Rank=("Rank", "first")
     )
```

```
    .reset_index()
)

player_week["sentiment_std"] = player_week["sentiment_std"].fillna(0)
```

From there, you can test whether the average sentiment for a player-week predicts fantasy points.

Testing for Position-Specific Effects

One reason the overall relationship may look weak is that sentiment may behave differently by position. Analysts may be more optimistic about tight ends, more cautious about running backs, or more reactive to quarterback play.

You can test that with interaction terms.

```
interaction_model = smf.ols(
   "TotalPoints ~ mean_sentiment * C(position)",
   data=player_week
).fit()

print(interaction_model.summary())
```

This is useful because it asks a more realistic question: not "does sentiment always predict performance?" but "does sentiment matter more for some positions than others?"

If the interaction terms are meaningful, then the overall average may have hidden the real pattern. If they are not, then sentiment may be mostly narrative framing rather than a predictive feature.

Bias Checks

1. Recency bias

   If analysts are mostly reacting to last week's performance, sentiment may be echoing recent results rather than predicting future ones. A simple way to begin checking this is to compare sentiment with previous-week points if you have that data available, or with current-week performance to see whether tone is backward-looking.
2. Coverage bias

   Some players get far more coverage than others. That can distort sentiment because star players are both discussed more and likely to receive more positive framing. If article_count matters, that suggests media attention itself may be part of the signal or part of the bias.
3. Analyst disagreement

   If articles conflict with each other, that disagreement may be informative. A high standard deviation could mean a player is volatile or hard to project.

This is a strong next step because it goes beyond simple positivity and asks whether volatility in analyst opinion predicts volatility in player output.

Checking Model Quality

Once you fit a model, you should not stop at the coefficient table. You also want to check whether the model is actually behaving well.

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10, 6))
sns.scatterplot(x=predictions, y=residuals, alpha=0.5)
plt.axhline(0, color="red", linestyle="--")
plt.title("Residual Plot")
plt.xlabel("Predicted Fantasy Points")
plt.ylabel("Residuals")
plt.show()
```

This kind of plot helps identify whether the model is systematically missing certain kinds of players, such as boom/bust options or low-volume players. If the residuals fan out or cluster, that is a sign the relationship is more complicated than a simple linear model can capture.

Next Steps

The biggest questions remain open: do these outlooks truly contain predictive value, or are they mostly recycling recent performance, biases, and narrative framing? When sentiment is positive, is it revealing insight, or simply reflecting what the market already believes? And when analysts disagree is that a hidden signal about upside and volatility?

You should now have all the tools necessary to dive deeper into the data and decide whether fantasy football article tone is real signal, or just better storytelling. I encourage you to also find additional sources of fantasy writeups to get an even broader look at the media sentiment. This could open the door to doing a comparison between predictive accuracy of different fantasy media outlets. There are many more paths to explore with this baseline dataset.